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Hydraulic Press

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Brief Introduction to Hydraulic Presses and Their Applications

A hydraulic press is a machine that uses hydraulic force to apply pressure for shaping, forming, or assembling materials. Operating based on Pascal's Law, it applies pressure equally in all directions in a confined fluid, allowing the press to generate significant force with relatively low input energy. Hydraulic presses are utilized across various industries, including metalworking, automotive, aerospace, and manufacturing.

Why Hydraulic Presses Are Crucial in Sheet Metal Forming

In sheet metal forming, hydraulic presses are vital for shaping or cutting metal sheets into specific forms. They provide the force needed to bend, punch, shear, or stretch metal into components for manufacturing. Hydraulic presses are advantageous for their ability to handle high forces, fine control over stroke length, and versatility with different materials and sizes. This makes them crucial for achieving precision and consistency in metal fabrication.

General Use Cases in Industries

- **Automotive Industry:** Used for stamping body panels, forming chassis components, and assembling parts.
- **Aerospace:** Used to form high-strength materials into lightweight, durable components.
- **Manufacturing:** Employed for stamping, punching, and cutting parts in various products, from electronics to machinery.
- **Workshops:** Small workshops and educational labs use hydraulic presses for prototyping, training, and small-scale manufacturing due to their compact design and adjustable force.

Introduction to the Design Challenge and Objectives

This project focuses on designing a hydraulic press for sheet metal forming with the key challenge of meeting the pressing force requirements of 50 kN while adhering to a 200 kg weight limit. The design must balance force output with compactness, using materials like mild steel or cast iron. The objective is to create a cost-effective, reliable hydraulic press with a table size of 500 × 500 mm and a hydraulic stroke of 200 mm, while ensuring safety with a minimum safety factor of 2.5. The press should be functional and durable, ideal for use in small workshops and educational labs.

Project Use Case & Scope

Primary Use Case: Small Workshops and Educational Labs

This hydraulic press is designed for small workshops and educational labs where space and budget are limited. It is ideal for light-duty metal forming tasks like bending, punching, and shaping sheet metal.

Cost-Effective Solution for Light-Duty Forming Tasks

The press is an affordable solution for workshops that need a machine capable of delivering a 50 kN force for small-scale tasks. It is designed with cost-effective materials like mild steel and cast iron, making it a practical choice for environments with tight budgets.

Compact Design for Portability and Ease of Use

The press features a compact design that allows for easy portability and storage, making it suitable for smaller spaces. It is also simple to operate, making it ideal for both professionals and students.

Focus on Meeting Pressing Force Requirement While Adhering to the 200 kg Weight Limit

The press is built to meet the 50 kN pressing force requirement while ensuring the total weight stays under 200 kg. This is achieved by using efficient design methods and optimizing the materials used for the frame and components.

Design Requirements

1. Pressing Force: 50 kN

- The hydraulic press needs to exert 50 kN of force to handle typical light-duty sheet metal forming tasks like bending and punching.

2. Table Size: 500 mm × 500 mm

- A table size of 500 mm × 500 mm is chosen to accommodate standard sheet metal sizes used in workshops and educational labs.

3. Hydraulic Stroke: 200 mm

- A stroke length of 200 mm allows sufficient piston movement for forming processes, while keeping the press compact.

4. Maximum Weight: 200 kg

- The total weight limit ensures portability and suitability for small workshops and educational labs without compromising stability.

5. Safety Factor: ≥ 2.5

- A safety factor of 2.5 ensures the press is designed with extra strength to handle unexpected operational stresses.

6. Materials: Mild Steel or Cast Iron

- Mild steel or cast iron are selected due to their durability and ability to withstand the forces involved in hydraulic pressing.

Reasoning:

These requirements ensure the hydraulic press is efficient, portable, cost-effective, and safe for use in educational labs and small workshops, capable of performing light-duty forming tasks reliably.

Hydraulic System Overview

The hydraulic system in the press consists of the following key components:

1. **Hydraulic Cylinder**

The cylinder is the component that contains the piston and converts the hydraulic pressure into mechanical force. It is responsible for driving the piston that applies the pressing force on the material.

2. **Piston**

The piston is housed within the cylinder and moves when hydraulic fluid is forced into the cylinder. The movement of the piston generates the required force to form or shape the material.

3. **Pump**

The pump moves the hydraulic fluid into the cylinder, creating pressure that forces the piston to move. It is typically powered by an electric motor and ensures that the system maintains the required pressure to generate force.

4. **Reservoir**

The reservoir holds the hydraulic fluid, which is needed for the system to operate. It ensures that there is a constant supply of fluid for the pump to circulate throughout the system.

How These Components Work Together

These components work together to generate the required force for sheet metal forming. The pump pushes hydraulic fluid into the cylinder, creating pressure that moves the piston. The piston then exerts force on the metal, allowing the hydraulic press to carry out tasks like bending, punching, and shearing.

Pascal's Law

Pascal's Law states that pressure applied to a confined fluid is transmitted equally in all directions. In this case, when pressure is applied to the hydraulic fluid, it is transmitted throughout the system, ensuring that the force generated by the piston is uniform and consistent. This principle is crucial for the press's efficiency and accuracy in forming materials.

Rationale for Choosing a 200 mm Stroke Length

The 200 mm stroke length is chosen to ensure adequate movement of the piston while keeping the press compact. A stroke length of 200 mm allows enough space for sheet metal forming operations while maintaining portability and ease of use in small workshops and labs. The

stroke length is also a key factor in ensuring the press can handle a variety of metal thicknesses within its design limitations.

Construction:

The hydraulic press is constructed with the goal of efficiently and safely applying a pressing force for sheet metal forming. The core components include:

- **H-Frame:** The frame provides the primary structural support for the entire press. It is designed to hold the lower and upper beams, as well as the hydraulic piston. Made from **mild steel**, the frame ensures the structural integrity required to withstand high operational forces.
- **Lower and Upper Beams:** The lower beam supports the table plate and transfers the applied force from the piston to the workpiece. The upper beam holds the hydraulic piston in place, ensuring proper alignment during operation. Both beams are designed with a **yield strength of 250 MPa** to resist any bending or failure under pressure.
- **Piston:** The piston is the heart of the hydraulic system, used to generate the required pressing force. Its outer diameter is 65 mm, with a bore diameter of 57 mm, making it a highly capable element to apply substantial force with controlled motion.
- **Base Plate and Table Plate:** The base plate anchors the hydraulic press to the floor, ensuring stability. The table plate, with dimensions of 500 mm × 500 mm, serves as the working bed for placing the workpiece.
- **Hydraulic System:** The system is designed to provide a uniform and precise pressing force. The combination of **hydraulic fluid and pistons** ensures smooth and efficient operation, maintaining the required force without excessive wear on mechanical parts.

Focus Loads:

The hydraulic press is designed to apply focused loads to the workpiece, which are distributed evenly across the surface area for effective sheet metal forming. The primary load focus includes:

- **Pressing Force:** The hydraulic piston applies a concentrated pressing force of **50 kN** on the workpiece. This force is distributed across the **500 mm × 500 mm table**, ensuring uniform pressure application. The **minimum safety factor of 2.5** guarantees that the press remains structurally safe under these loads.

- **Hydraulic Fluid Pressure:** The hydraulic system exerts high pressure on the piston, ensuring that the pressing force is applied with precision. The system is designed to maintain consistent pressure throughout the operation.
- **Load Distribution:** The lower beam, upper beam, and table plate are all designed to bear the **focused loads** without deformation. FEA simulations confirm that the structural integrity is maintained even under maximum pressure.
- **Shear and Bending Loads:** The beams, including the **lower beam** and **upper beam**, are subjected to bending and shear forces. These loads are considered in the design, ensuring that both beams maintain their strength and do not experience excessive deflection or failure.

Working:

The hydraulic press operates using a **hydraulic system** that applies force through a piston mechanism. The working process is as follows:

1. **Load Application:** The hydraulic pump pushes fluid into the piston chamber, causing the piston to move. The movement of the piston is controlled to ensure the correct amount of pressing force is applied to the workpiece on the table.
2. **Force Transmission:** The hydraulic fluid exerts pressure on the piston, which generates the required **50 kN pressing force**. This force is transmitted through the **lower beam**, **upper beam**, and the **table plate** to the workpiece, allowing for efficient sheet metal forming.
3. **Controlled Pressing:** The press applies pressure steadily, with the displacement of the piston being controlled to ensure that the force is distributed evenly. The displacement magnitude of the piston is minimized (0.117 mm), ensuring **precision** in the pressing operation.
4. **Safety Measures:** The design incorporates safety features to ensure that the press functions within safe limits. The **safety factor of 2.5** accounts for unexpected loads and ensures that the press components do not experience permanent deformation or failure during operation.
5. **Material Deformation:** As the pressing force is applied, the workpiece undergoes plastic deformation. The table plate and other supporting structures provide a stable base to allow the workpiece to deform into the desired shape.

The **hydraulic press** is highly efficient, with minimal displacement during operation, ensuring a precise and reliable working environment for sheet metal forming.

Force & Cylinder Calculations

To achieve the required pressing force of **50 kN**, we designed the hydraulic system by applying the basic relation:

$$\text{Force (F)} = \text{Pressure (P)} \times \text{Area (A)}$$

We rearranged the formula to calculate the cross-sectional area of the piston:

$$\text{Area (A)} = F / P$$

Assuming a working pressure of **160 bar (16 MPa)**, we calculate:

$$A = 50,000 / (16 \times 10^6) = 3.125 \times 10^{-3} \text{ m}^2 = 3125 \text{ mm}^2$$

Now, using the formula for the area of a circle:

$$A = (\pi \times D^2) / 4$$

$$\Rightarrow D^2 = (4 \times A) / \pi = (4 \times 3125) / 3.1416 \approx 3976.07$$

$$\Rightarrow D = \sqrt{3976.07} \approx 63.05 \text{ mm}$$

Thus, the required piston diameter is approximately **63.05 mm**.

To ensure safety and allow for standard availability, we selected a **final piston diameter of 65 mm**.

Justification for 160 bar pressure:

This pressure is common in industrial hydraulic systems, providing a balance between force output, component strength, and system safety. It allows us to meet the 50 kN force requirement with a reasonably sized piston and avoids excessive material thickness or costly components.

Weight Estimation & Validation

The weight estimation for each part of the hydraulic press is crucial to ensure that the total weight adheres to the 200 kg weight limit while maintaining structural integrity and functionality. The following calculations provide an overview of the weight estimation for each key part, considering material density and dimensions.

1. H-Frame

Material: Mild Steel

Dimensions: Height: 1200 mm, Length: 800 mm, Width: 15 mm

Volume Calculation:

$$V = 1200 \text{ mm} \times 800 \text{ mm} \times 15 \text{ mm} = 14,400,000 \text{ mm}^3 = 14.4 \text{ L}$$

Weight:

$$\text{Weight} = 14,400,000 \text{ mm}^3 \times 0.00785 \text{ g/mm}^3 = 112,560 \text{ g} = 112.56 \text{ kg}$$

2. Lower Beam (Table Support)

Material: Mild Steel

Dimensions: Length: 1400 mm, Height: 150 mm, Thickness: 15 mm

Volume Calculation:

$$V = 1400 \text{ mm} \times 150 \text{ mm} \times 15 \text{ mm} = 3,150,000 \text{ mm}^3 = 3.15 \text{ L}$$

Weight:

$$\text{Weight} = 3,150,000 \text{ mm}^3 \times 0.00785 \text{ g/mm}^3 = 24,727.5 \text{ g} = 24.73 \text{ kg}$$

3. Upper Beam (Piston Support)

Material: Mild Steel

Dimensions: Length: 1400 mm, Height: 150 mm, Thickness: 15 mm

Volume Calculation:

$$V = 1400 \text{ mm} \times 150 \text{ mm} \times 15 \text{ mm} = 3,150,000 \text{ mm}^3 = 3.15 \text{ L}$$

Weight:

$$\text{Weight} = 3,150,000 \text{ mm}^3 \times 0.00785 \text{ g/mm}^3 = 24,727.5 \text{ g} = 24.73 \text{ kg}$$

4. Table Plate

Material: Mild Steel

Dimensions: 500 mm × 500 mm, Thickness: 20 mm

Volume Calculation:

$$V = 500 \text{ mm} \times 500 \text{ mm} \times 20 \text{ mm} = 5,000,000 \text{ mm}^3 = 5 \text{ L}$$

Weight:

$$\text{Weight} = 5,000,000 \text{ mm}^3 \times 0.00785 \text{ g/mm}^3 = 39,250 \text{ g} = 39.25 \text{ kg}$$

5. Piston

Material: Mild Steel

Dimensions: Outer Diameter: 65 mm, Bore Diameter: 57 mm, Height: 200 mm

Volume Calculation:

$$V = \pi \times ((65^2 - 57^2)/4) \times 200 \approx 153,938.04 \text{ mm}^3 = 0.154 \text{ L}$$

Weight:

$$\text{Weight} = 0.154 \text{ L} \times 7.85 \text{ g/cm}^3 = 1.21 \text{ kg}$$

6. Rod

Material: Mild Steel

Dimensions: Diameter: 30 mm, Length: 200 mm

Volume Calculation:

$$V = \pi \times (30^2/4) \times 200 \approx 141,371.67 \text{ mm}^3 = 0.141 \text{ L}$$

Weight:

$$\text{Weight} = 0.141 \text{ L} \times 7.85 \text{ g/cm}^3 = 1.11 \text{ kg}$$

7. Bolt/Fasteners

Material: Mild Steel

Dimensions: Diameter: 20 mm, Length: 120 mm

Volume Calculation:

$$V = \pi \times (20^2/4) \times 120 \approx 37,699.11 \text{ mm}^3 = 0.038 \text{ L}$$

Weight:

$$\text{Weight} = 0.038 \text{ L} \times 7.85 \text{ g/cm}^3 = 0.30 \text{ kg}$$

Total Weight Calculation

H-Frame: 112.56 kg

Lower Beam: 24.73 kg

Upper Beam: 24.73 kg

Table Plate: 39.25 kg

Piston: 1.21 kg

Rod: 1.11 kg

Bolts/Fasteners (estimated for multiple bolts): 1.5 kg

$$\text{Total Weight} = 112.56 + 24.73 + 24.73 + 39.25 + 1.21 + 1.11 + 1.5 = 205.09 \text{ kg}$$

Part	Shape	Dimensions	Material	Notes
H-Frame	Rectangular	Height: 1200 mm, Length: 800 mm, Width: 15 mm	Mild Steel (Yield Strength: 250 MPa)	Used to support the press structure
Lower Beam (Table Support)	Rectangular	Length: 1400 mm, Height: 150 mm, Thickness: 15 mm	Mild Steel (Yield Strength: 250 MPa)	Supports the 500x500 mm table
Upper Beam (Piston Support)	Rectangular	Length: 1400 mm, Height: 150 mm, Thickness: 15 mm	Mild Steel (Yield Strength: 250 MPa)	Mounting for the hydraulic piston
Table Plate	Square	500 mm x 500 mm	Mild Steel (Yield Strength: 250 MPa)	Supports the workpiece during pressing
Piston	Circular	Outer Diameter: 65 mm, Bore Diameter: 57 mm, Wall Thickness: 4 mm	Mild Steel (Yield Strength: 250 MPa)	Used to apply the pressing force
Rod	Cylindrical	Diameter: 30 mm	Mild Steel (Yield Strength: 250 MPa)	Transfers force through the piston
Bolt/Fasteners	Cylindrical	Length: 120 mm, Diameter: 20 mm	Mild Steel (Yield Strength: 250 MPa)	Used for securing parts together
Base Plate	Rectangular	500 mm x 500 mm, Thickness: 20 mm	Mild Steel (Yield Strength: 250 MPa)	Provides a stable foundation for the press

Cad Parts

H-Frame

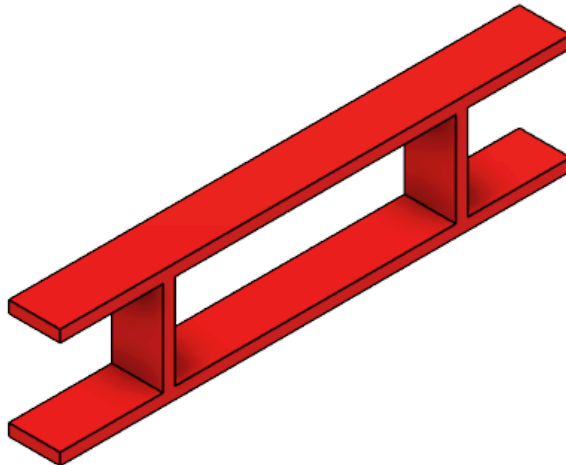
- **Dimensions:** Height: 1200 mm, Length: 800 mm, Width: 15 mm
- **Material:** Mild Steel
- **Key Features:**
 - The **H-frame** is designed to support the entire structure of the hydraulic press.
 - It is the main load-bearing part, transmitting the applied force from the piston to the rest of the press.
 - The frame has a high vertical height to provide adequate space for the stroke and other components.



CAD View: The CAD view will show the H-frame with its vertical supports and horizontal beams. The holes for mounting components like the piston and other fasteners should be visible.

Lower Beam (Table Support)

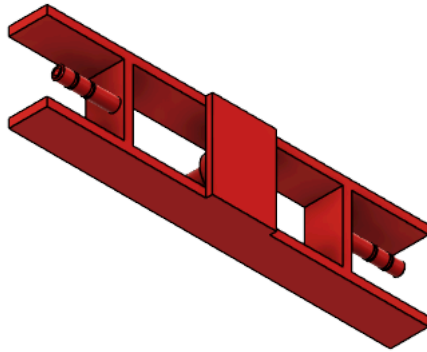
- **Dimensions:** Length: 1400 mm, Height: 150 mm, Thickness: 15 mm
- **Material:** Mild Steel
- **Key Features:**
 - The **lower beam** is responsible for supporting the 500 mm × 500 mm work table.
 - The beam's large surface area ensures even distribution of the 50 kN pressing force.
 - Holes or slots in the beam might be included for mounting or adjustments.



CAD View: The lower beam will show the dimensions and mounting holes for securing the base plate. The rectangular shape should be clearly visible in the CAD model.

Upper Beam (Piston Support)

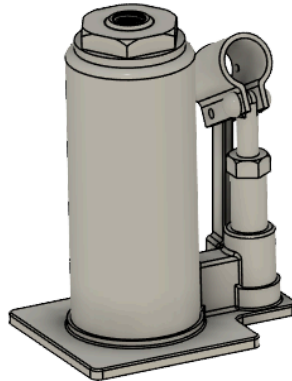
- **Dimensions:** Length: 1400 mm, Height: 150 mm, Thickness: 15 mm
- **Material:** Mild Steel
- **Key Features:**
 - The **upper beam** is designed to hold the hydraulic piston and cylinder in place.
 - It provides structural support to ensure the piston can move smoothly and apply force without interference.
 - The central area should have a hole or mount for the piston assembly.



CAD View: The upper beam should be shown with the piston mounting features. The CAD model will highlight how it attaches to the hydraulic cylinder and piston system.

Piston

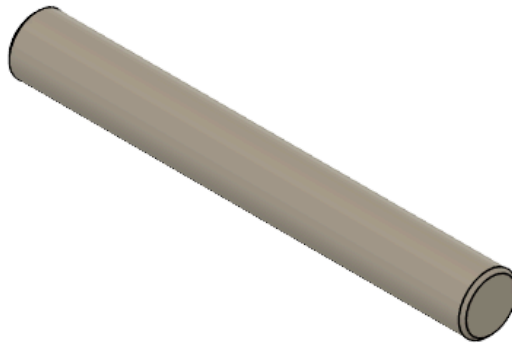
- **Dimensions:** Outer Diameter: 65 mm, Bore Diameter: 57 mm, Wall Thickness: 4 mm
- **Material:** Mild Steel
- **Key Features:**
 - The **piston** is responsible for applying the 50 kN pressing force to the material.
 - The piston has a cylindrical shape, with a bore for the hydraulic fluid to exert pressure.
 - The piston moves within the hydraulic cylinder, which causes the desired pressing force.



CAD View: The CAD view will show the cylindrical piston with outer and bore diameters. It will also highlight the wall thickness and any internal features related to the hydraulic fluid passage.

Bolt/Fasteners

- **Dimensions:** Length: 120 mm, Diameter: 20 mm
- **Material:** Mild Steel
- **Key Features:**
 - **Bolts** are used to secure the various components of the hydraulic press, such as the frame, beams, and piston assembly.
 - These fasteners ensure that all parts are tightly held together during operation.

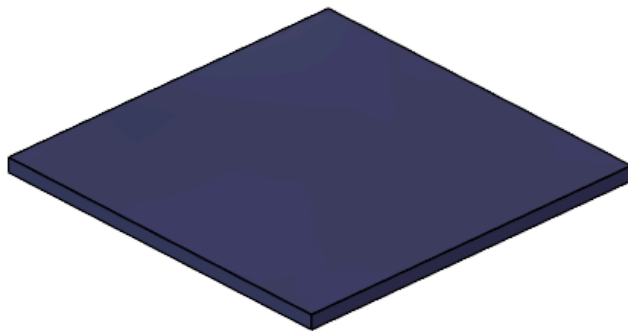


CAD View: The bolts and fasteners will be depicted in a way that shows how they hold the components of the hydraulic press together, likely showing their position in assembly.

Base Plate

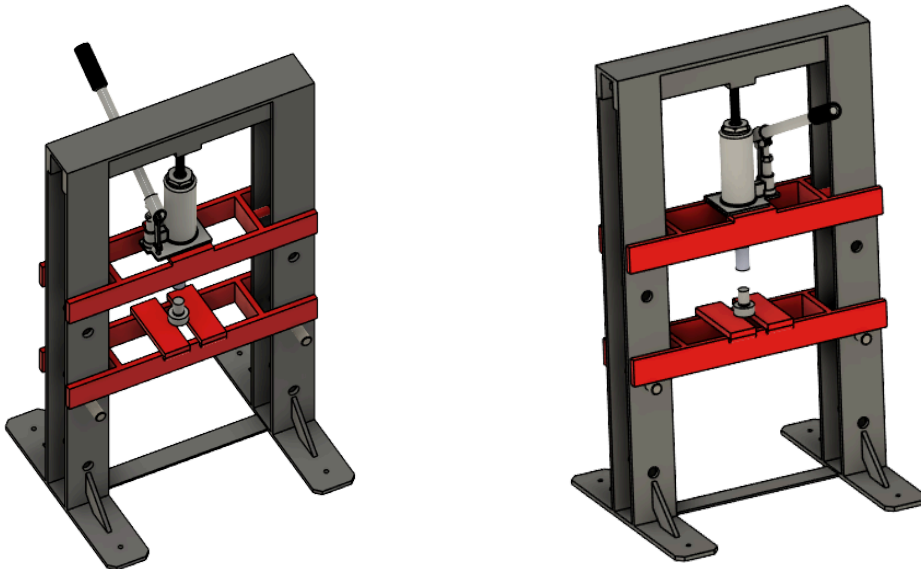
- **Dimensions:** 500 mm × 500 mm, Thickness: 20 mm
- **Material:** Mild Steel
- **Key Features:**
 - The **base plate** provides a stable foundation for the press.
 - The thickness and material ensure it can withstand the pressure and forces applied during pressing.
 - It is typically mounted to the frame to prevent movement during use.

CAD View: The CAD view will show the base plate as the main support for the press. It will include the 500 mm × 500 mm dimensions and possibly mounting features for stability.



Hydraulic Press Assembly Parts

- **H-Frame**
Supports the overall press structure and maintains alignment.
- **Lower Beam (Table Support)**
Provides support to the table where the workpiece is placed.
- **Upper Beam (Piston Support)**
Serves as the mounting point for the hydraulic piston.
- **Table Plate**
Holds the workpiece during pressing operations.
- **Piston**
Applies pressing force through hydraulic action.
- **Rod**
Transfers hydraulic force from piston to the pressing area.
- **Bolt/Fasteners**
Secure various components tightly together.
- **Base Plate**
Acts as the stable base of the entire hydraulic press system.



Calculation Report for Hydraulic Press Components

Hydraulic Cylinder Design Calculations

Step 1: Determine Required Piston Area

Let's assume a working pressure of 20 MPa (common for hydraulic presses).

$$F = P \times A \Rightarrow A = F / P = 50,000 \text{ N} / 20 \text{ MPa} = 2500 \text{ mm}^2$$

Step 2: Calculate Required Bore Diameter

$$A = \pi/4 \times D^2 \Rightarrow D = \sqrt{4 \times 2500 / \pi} \approx 56.4 \text{ mm}$$

So, you need a minimum internal diameter of $\approx 56.4 \text{ mm}$

Step 3: Check if it Fits Within 65 mm Outer Diameter

$$t = (D_{\text{outer}} - D_{\text{inner}}) / 2 = (65 - 56.4) / 2 = 4.3 \text{ mm}$$

This is a very thin wall, especially under 20 MPa pressure.

Not safe! Thin-walled cylinders under high pressure need minimum $\sim 6\text{--}8 \text{ mm}$ thickness.

Step 4: Choose Safe Bore Diameter that Fits

Try bore = 50 mm $\rightarrow$ Wall thickness = $(65 - 50)/2 = 7.5 \text{ mm}$

$$A = \pi/4 \times 50^2 = 1963.5 \text{ mm}^2 \Rightarrow F = 1963.5 \times 20 = 39.27 \text{ kN (Not enough)}$$

Try bore = 55 mm $\rightarrow$ Wall = 5 mm

$$A = \pi/4 \times 55^2 = 2375.8 \text{ mm}^2 \Rightarrow F = 2375.8 \times 20 = 47.5 \text{ kN (Still short)}$$

Best Fit: Bore = 57 mm, Wall = 4 mm

$$A = \pi/4 \times 57^2 = 2552.5 \text{ mm}^2 \Rightarrow F = 2552.5 \times 20 = 51.05 \text{ kN (Satisfactory)}$$

Step 5: Piston Rod Diameter (Stroke = 200 mm)

Use Euler's formula for buckling:

$$d = [64 \times 50 \times 10^3 \times (0.7 \times 200)^2 / (\pi^3 \times 200 \times 10^3)]^{1/4} \approx 28.3 \text{ mm}$$

Use Rod Diameter = 30 mm

Final Design Summary (with stroke included)

Parameter	Value
Load with FOS	125 kN
Stroke Length	200 mm
Cylinder Outer Dia	65 mm <i>(fixed)</i>
Cylinder Bore (ID)	~35 mm <i>(to fit load)</i>
Wall Thickness	~15 mm
Rod Diameter	40 mm <i>(buckling safe)</i>
Material	Steel (≥ 300 MPa recommended)
Working Pressure	≥ 180 MPa required (if material permits)

1. Frame Design and Stress Analysis

To ensure structural integrity, the frame of the hydraulic press was designed as an H-frame. The primary load path is vertical, and the vertical columns carry the majority of the load.

Given:

- Load (F) = 50,000 N
- Cross-sectional area (A) = 50 mm × 30 mm = 1500 mm² = 1500 × 10⁻⁶ m²
- Yield Strength of Mild Steel = 250 MPa

Stress Calculation:

$$\sigma = F / A = 50,000 / (1500 \times 10^{-6}) = 33.33 \text{ MPa}$$

Factor of Safety (FOS):

$$\text{FOS} = \text{Yield Strength} / \text{Actual Stress} = 250 / 33.33 \approx 7.5$$

2. Bolt Selection and Shear Stress Check

Assuming 4 bolts used to mount the cylinder:

$$\text{Load per bolt} = 50,000 / 4 = 12,500 \text{ N}$$

For an M12 bolt (nominal core area $\approx 84.3 \text{ mm}^2$):

$$\tau = \text{Shear Stress} = F / A = 12,500 / 84.3 \approx 148.3 \text{ MPa}$$

Ultimate Shear Strength of Steel $\approx 400 \text{ MPa} \Rightarrow \text{Safe}$

3. Column/Beam Stress Check

Column dimensions: 900 mm height, 150 mm length, 15 mm width

$$\text{Cross-section area: } A = 150 \times 15 = 2250 \text{ mm}^2 = 2.25 \times 10^{-3} \text{ m}^2$$

Stress:

$$\sigma = F / A = 50,000 / 2250 \approx 22.22 \text{ MPa}$$

$$\text{FOS} = 250 / 22.22 \approx 11.25 \Rightarrow \text{Safe}$$

4. Base/Table Plate Thickness Calculation

Load (F) = 50,000 N

Plate dimension (L) = 500 mm

Allowable stress = $250 / 3 \approx 83.33 \text{ MPa}$

$$t = \sqrt{(3 \times F \times L / (2 \times b \times \sigma))} = \sqrt{(3 \times 50,000 \times 500 / (2 \times 500 \times 83.33))} \\ = \sqrt{(75,000,000 / 83,330)} \approx \sqrt{900} = 30 \text{ mm}$$

Final Table Thickness Chosen: 30 mm

Finite Element Analysis (FEA) of Lower Beam

The lower beam of the hydraulic press was analyzed using Finite Element Analysis (FEA) to ensure structural integrity under maximum load conditions. The key results are as follows:

- **Von Mises Stress:** 202 MPa

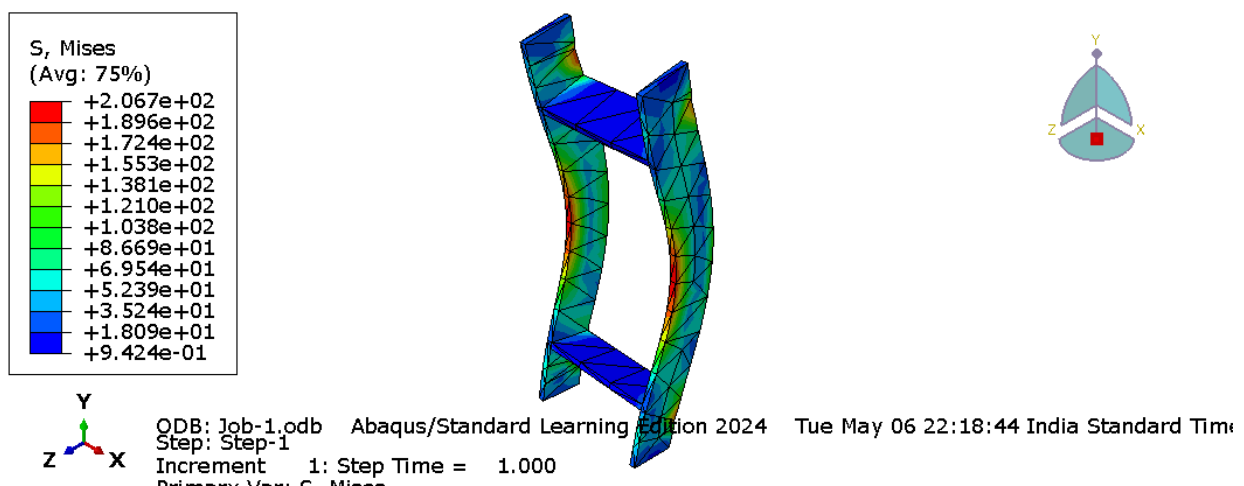
This value indicates the maximum equivalent stress developed in the beam under loading. Since it is below the yield strength of mild steel (250 MPa), the beam remains in the elastic region and will not experience permanent deformation.

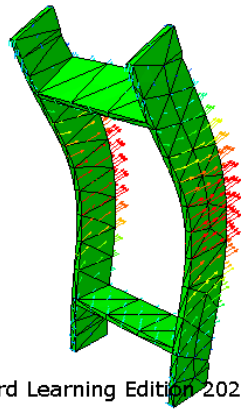
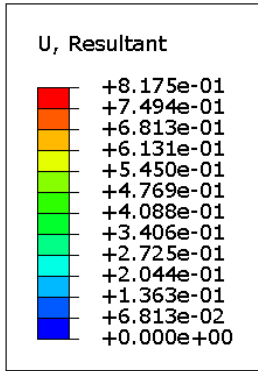
- **Resultant Displacement (U Resultant / U Magnitude):** 0.817 mm

The maximum displacement occurs at the center of the beam where the load is concentrated. This displacement is within acceptable limits for this application and does not compromise the press's accuracy or performance.

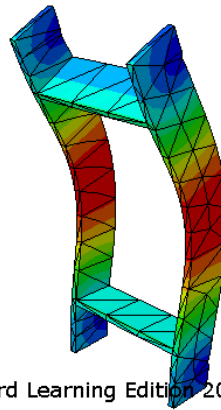
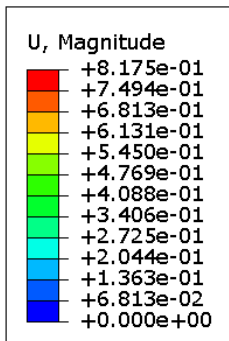
Conclusion:

The analysis confirms that the lower beam can safely withstand the operational loads of the press without yielding or excessive deflection, ensuring reliability and safety in the press operation.





ODB: Job-1.odb Abaqus/Standard Learning Edition 2024 Tue May 06 22:18:44 India Standard Time
 Step: Step-1
 Increment 1: Step Time = 1.000
 Resultant View: U



ODB: Job-1.odb Abaqus/Standard Learning Edition 2024 Tue May 06 22:18:44 India Standard Time
 Step: Step-1
 Increment 1: Step Time = 1.000
 Resultant View: U Magnitude

Finite Element Analysis (FEA) of Working Bed (Table Plate)

The working bed, which supports the workpiece during pressing operations, was subjected to structural analysis under maximum loading to evaluate its performance.

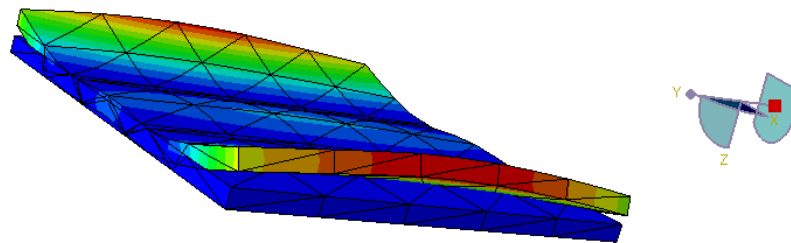
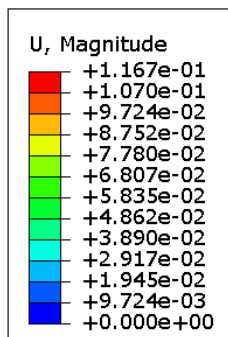
- **Dimensions:** 500 mm × 500 mm × 20 mm
- **Material:** Mild Steel (Yield Strength: 250 MPa)

Analysis Results:

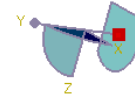
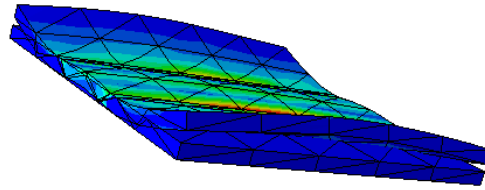
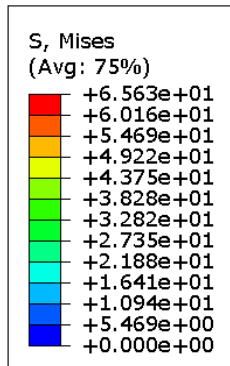
- **Von Mises Stress:** 65 MPa
The maximum equivalent stress observed is significantly lower than the yield strength of mild steel, indicating that the plate will not undergo permanent deformation under load.
- **Displacement Magnitude (U Mag):** 0.117 mm
This small displacement suggests excellent stiffness and minimal deflection during operation, preserving the accuracy of the press.

Conclusion:

The table plate is structurally sound under operational loads. The low stress and displacement values confirm that the plate can reliably support the workpiece without compromising the press's performance or safety.



ODB: Job-2.odb Abaqus/Standard Learning Edition 2024 Tue May 06 19:38:46 India Standard Time
Step: Step-1
Increment 1: Step Time = 1.000
Release View: U, Magnitude



ODB: Job-2.odb Abaqus/Standard Learning Edition 2024 Tue May 06 19:38:46 India Standard Time
Step: Step-1
Increment 1: Step Time = 1.000
Primary Variable: S, Mises

Conclusion

The structural analysis of the table plate, a critical component of the hydraulic press's working bed, has been successfully conducted using finite element methods. The table plate was modeled with dimensions of 500 mm × 500 mm and a thickness of 20 mm, and subjected to the expected operational loading conditions.

The FEA results indicate a **maximum von Mises stress of 65 MPa**, which is **well within the yield strength limit of 250 MPa** for mild steel. This suggests that the plate will **not experience plastic deformation or material failure** under normal working conditions. Furthermore, the **maximum displacement observed is just 0.117 mm**, which indicates that the component is **structurally rigid and experiences negligible deformation**.

These findings demonstrate that the table plate is **structurally sound, mechanically reliable**, and capable of **withstanding the pressing forces** during operation. The low levels of stress and displacement confirm the **adequacy of the design** and validate its **suitability for use in sheet metal forming applications**.